Impact of Equivariance in Transformer-Based PDE Learning

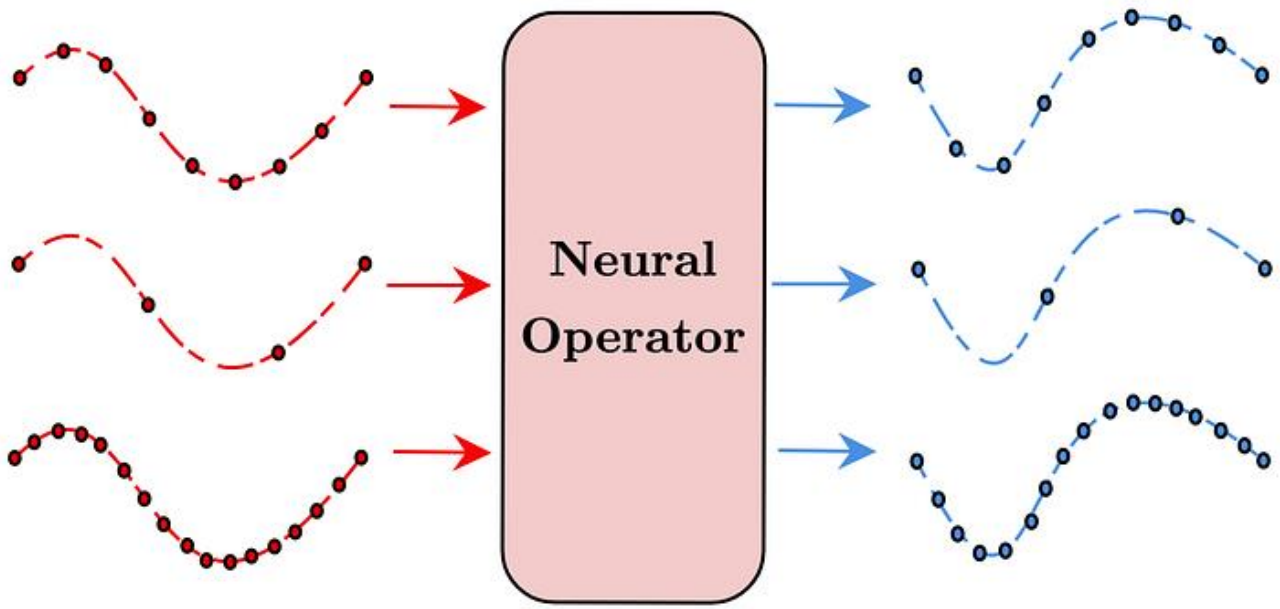
Mathematical Structures in Deep Neural Networks

2026-03-18

Jack Elliot Collier Ryder

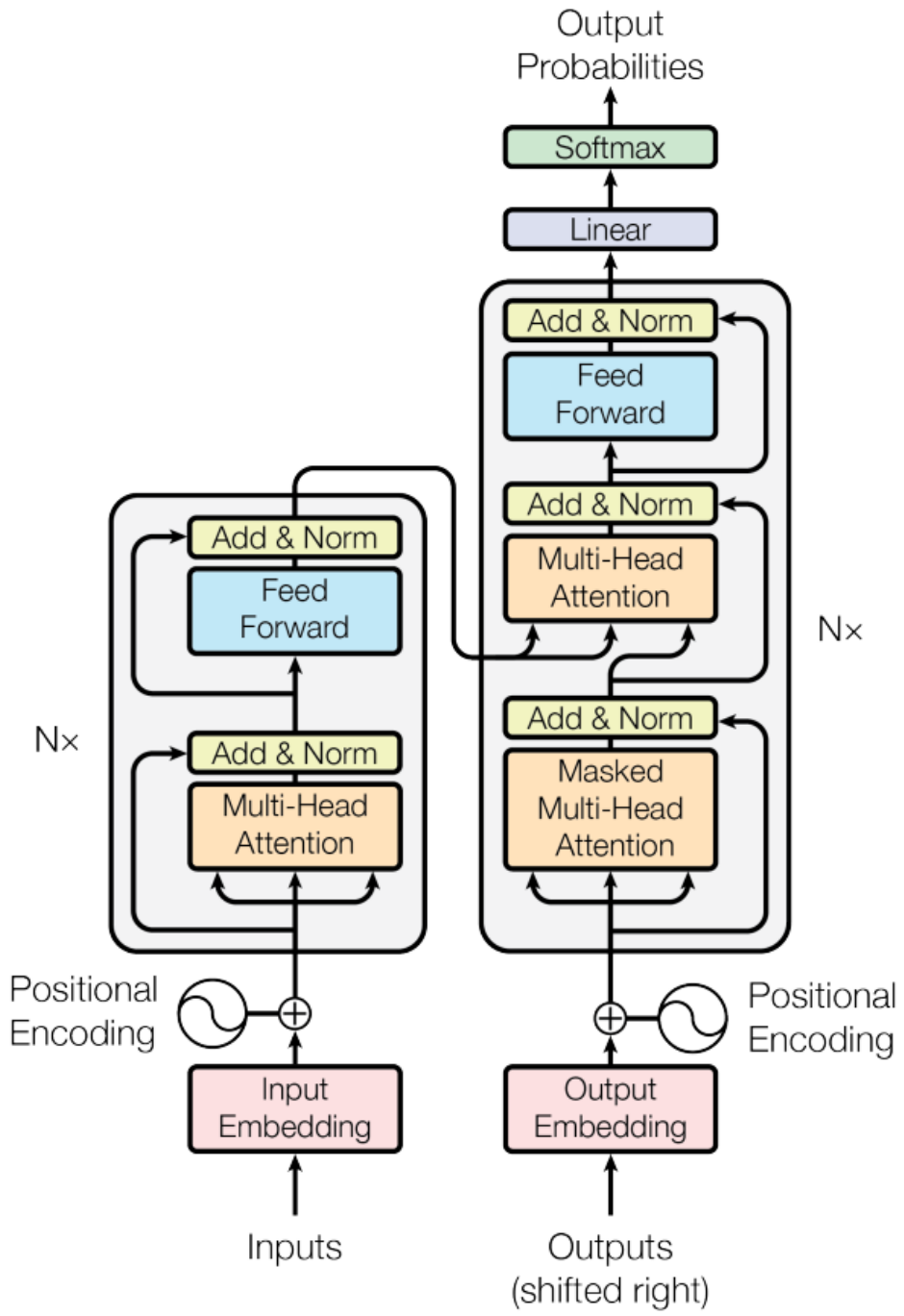
Operator Learning

- Mappings between function spaces
- PDE surrogates
- Several architectures:
  - Fourier Neural Operators
  - U-Net
  - Transformers

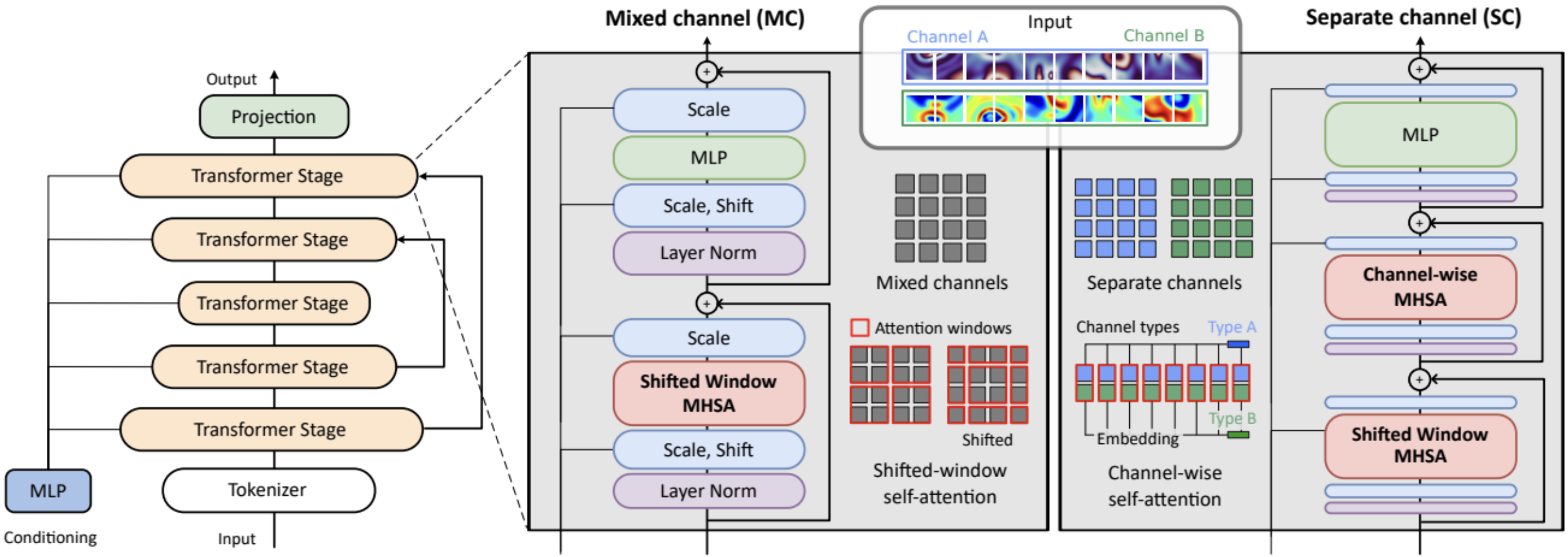


Transformer

- Attention is All You Need – 2017
- Natural Language Processing
- Vision, Diffusion, Dance
- Operator Learning?



PDE-Transformer



Vanilla Group Equivariant Vision Transformer: Simple and Effective

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Abstract

Incorporating symmetry priors as inductive biases to design equivariant Vision Transformers (ViTs) has emerged as a promising avenue for enhancing their performance. However, existing equivariant ViTs often struggle to balance performance with equivariance, primarily due to the challenge of achieving holistic equivariant modifications across the diverse modules in ViTs—particularly in harmonizing the Self-Attention mechanism with Patch Embedding. To address this, we propose a straightforward framework that systematically renders key ViT components, including patch embedding, self-attention, and positional encodings, and Down/Up-Sampling, equivariant, thereby constructing ViTs with guaranteed equivariance. The resulting architecture serves as a plug-and-play replacement that is both theoretically grounded and practically versatile, scaling seamlessly even to Swin Transformers. Extensive experiments demonstrate that our equivariant ViTs consistently improve performance and data efficiency across a wide spectrum of vision tasks.

der the aforementioned transformations. However, standard Vision Transformers (ViTs) have not explicitly embedded such inductive biases into their design. Consequently, current ViT models typically rely on massive datasets and extensive data augmentation to implicitly learn these symmetries—a process that demands substantial data and computational resources while still failing to guarantee the systematic embedding of symmetry priors [14, 16, 35].

It is noteworthy that in recent years, the emergence of equivariant CNN architectures [8, 38, 53] has achieved remarkable success in embedding symmetry priors, demonstrating that directly designing equivariant model architectures offers unique advantages in parameter efficiency and performance enhancement, compared to learning symmetries from training data. This insight inspires the recognition that designing guaranteed equivariant architecture for ViTs is a highly promising research direction.

However, replicating the successful design of equivariant architectures from CNNs to Vision Transformers (ViT) has been proven to be challenging. Most existing approaches leverage the intrinsic permutation equivariance of

Equivariant Transformer

- Patch embedding replaced by lifting convolution
- Linear layers replaced by group-mixing equivariant linear maps
- Self-attention is performed on resulting fibered representation
- Relative positional bias is canonicalized under the group action
- LayerNorm shares affine parameters across fibers
- Down/up-sampling: canonicalize -> resize -> transform back

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Equivariant Transformer

- Started from official PDE-Transformer and kept the same hierarchical encoder-decoder structure
- Replaced the transformer core by equivariant patch embedding, Q/K/V projections, MLPs, positional bias, and normalization
- Converted conditioning to act consistently across fibers
- Canonicalized original PDE-Transformer resizers
- Kept original output format, but replaced final projection by an equivariant head

Shallow Water Equations

- PDE Bench
- D4



Fairness Comparison

- Parameter-matching
  - Vanilla paper
- Compute matching
- Wall-clock matching

On Fairly Comparing Group Equivariant Convolutional Networks

Lucas Roos [1] **Steve Kroon** [1 2]

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Abstract

This paper investigates the flexibility of Group Equivariant Convolutional Neural Networks (G-CNNs), which specialize conventional neural networks by encoding equivariance to group transformations. Inspired by splines, we propose new metrics to assess the complexity of ReLU networks and use them to quantify and compare the flexibility of networks equivariant to different groups. Our analysis suggests that the current practice of comparing networks by fixing the number of trainable parameters unfairly affords models equivariant to larger groups additional expressivity. Instead, we advocate for comparisons based on a fixed computational budget—which we empirically show results in more similar levels of network flexibility. This approach allows one to better disentangle the impact of constraining networks to be equivariant from the increased expressivity they are typically granted in the literature, enabling one to obtain a more nuanced view of the impact of enforcing equivariance. Interestingly, our experiments indicate that enforcing equivariance results in *more* complex fitted functions even when controlling for compute, despite reducing network expressivity.

interest to the deep learning community: features are semantically similar regardless of their location in the image. Instead of learning this symmetry from the training data, convolutional layers in CNNs are constrained a priori to be *equivariant* to translations. Group equivariant convolutional neural networks (G-CNNs) (Cohen & Welling, 2016) provide a framework for generalizing translational convolutions, allowing for the creation of architectures that are equivariant to other group transformations. G-CNNs have shown to be beneficial for a variety of tasks exhibiting symmetry, both theoretically (Kondor & Trivedi, 2018; Elesedy & Zaidi, 2021) and empirically (Walters et al., 2021; Dey et al., 2021; Lafarge et al., 2021; Bekkers et al., 2018; Zhu et al., 2022). G-CNNs are straightforward to implement for discrete groups and have also shown improvements over regular CNNs on more traditional image recognition tasks (Cohen & Welling, 2016; Weiler & Cesa, 2019; Hoogeboom et al., 2018). However, they have yet to enter mainstream use, with most practitioners still employing regular CNNs.

One drawback of incorporating group equivariance into architectures, potentially acting as a barrier to mainstream adoption, is the added computational expense: when the number of trainable parameters in a network is fixed, the computational cost of applying a group convolution scales linearly with the group size. In order to demonstrate the efficacy of their models, many works (Cohen & Welling, 2016; Knigge et al., 2022; Hoogeboom et al., 2018; Weiler

One-Step Training

- 10 epochs, 128x128, 100 time steps, 1000 samples

- Cosine decay from 10^{-3}

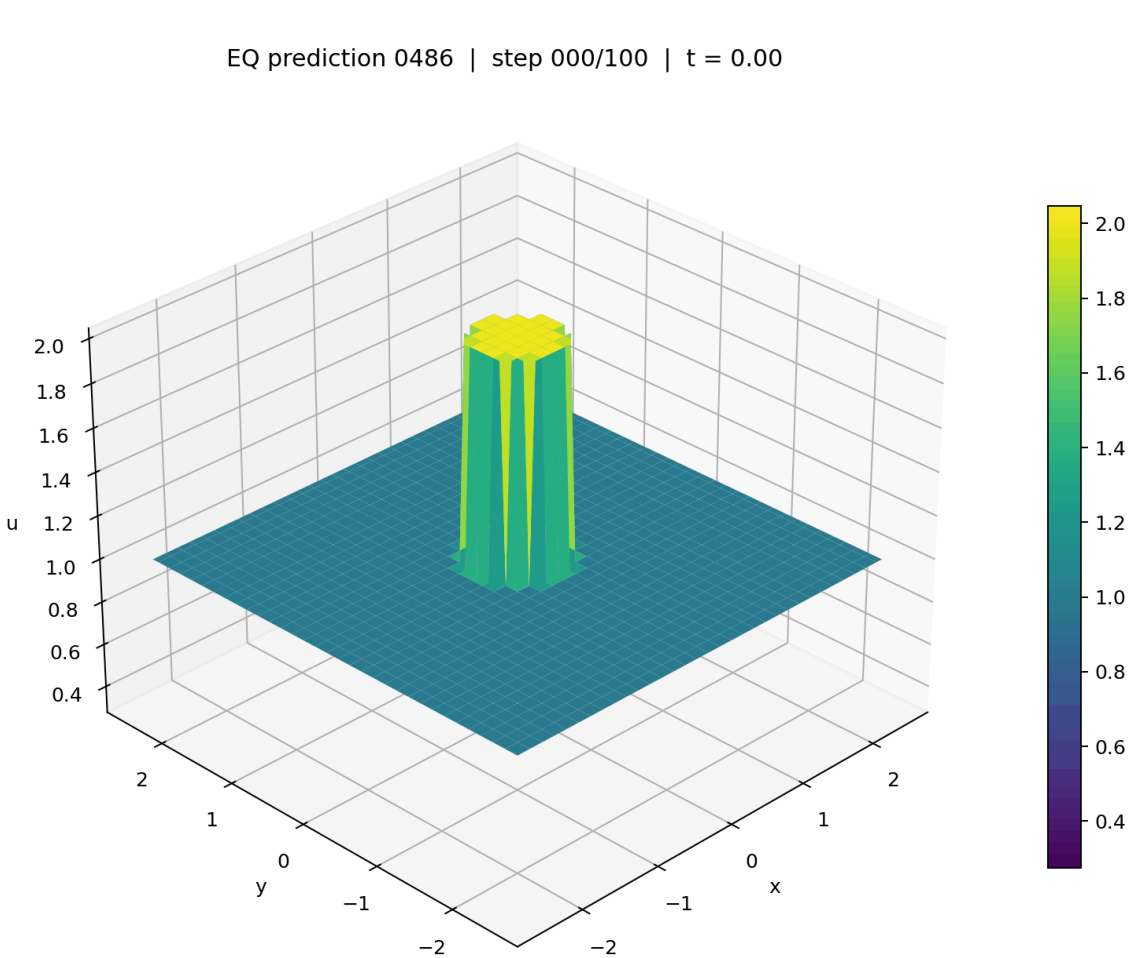
Model	Params	RMSE	D4 error	Train time	Inference time
Baseline	3.08M	0.00387	0.00124	**1h 34m**	*2.01 ms*
Baseline + Avg.	3.08M	0.00286	$\mathbf{3.61 \cdot 10^{-8}}$	–	14.25 ms
Augmentation	3.08M	0.00350	0.00105	*1h 37m*	**1.77 ms**
Regularization	3.08M	0.00319	0.00080	2h 46m	**1.77 ms**
Manifest EQ	2.98M	*0.00259*	$1.07 \cdot 10^{-7}$	5h 16m	7.20 ms
Baseline 112 (width-matched)	22.86M	0.00575	0.00224	2h 53m	–
Baseline (34 ep.)	3.08M	**0.00159**	0.00057	4h 45m	*2.01 ms*

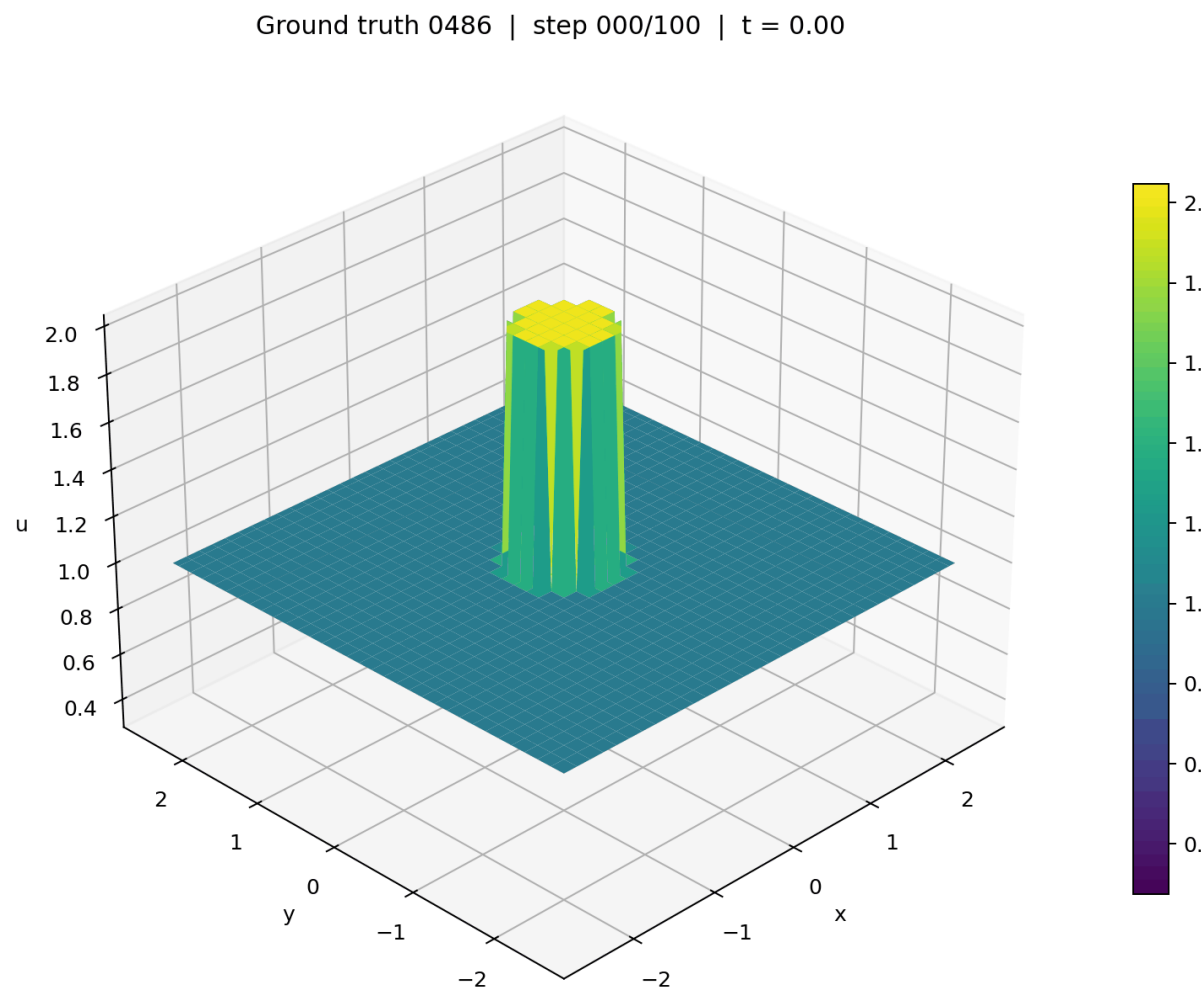
Rollout-Based Training

- 5 epochs, Linear warmup + Cosine decay from 10^{-4}

- Pushforward hybrid loss

Model	RMSE	Rollout RMSE	Train time
Baseline 48	0.00263	0.00688	**1h 58m**
Baseline 112	**0.00160**	**0.00372**	*3h 45m*
Manifest EQ	*0.00174*	*0.00622*	6h 49m


EQ prediction 0486 | step 000/100 | t = 0.00
u
x
y


Ground truth 0486 | step 000/100 | t = 0.00
u
x
y

Conclusions

- Equivariance remains interesting topic of study in PDE-based operator learning

- Scale up!
  - More datasets
  - More channels
  - Bigger models
  - More symmetries